

TRAFFIC INTELLIGENCE COMMAND

SYS.ID // PRK-INTEL-V1.0

STATUS : ● ONLINE

HACKATHON SUBMISSION / JURY REVIEW

AI-Driven Parking Intelligence for Targeted Enforcement.

A hotspot-ranking, forecasting and prescriptive enforcement engine —
built on raw parking-violation data, road network intelligence and predictive modeling.

RECORDS PROCESSED

298,450

PIPELINE PHASES

8 + 4 Layers

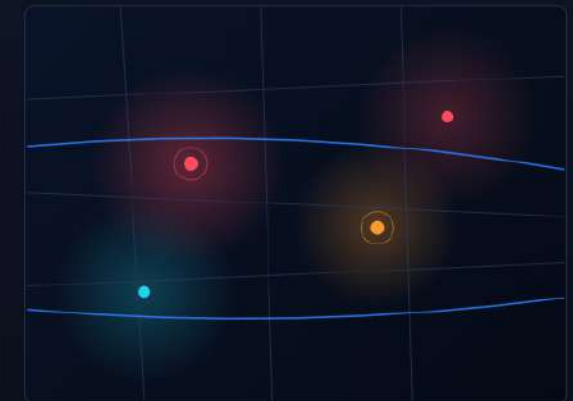
OUTPUT MODE

Prescriptive

PRESENTED BY **TEAM INNOVATORS**

LIVE OPS // CITY GRID

12:48:07



ACTIVE HOTSPOTS

147

EMERGING ZONES

23 ↑

CHRONIC OFFENDERS

412

PATROLS DISPATCHED

28 / 32

CONFIDENTIAL • JURY REVIEW EDITION

THE PROBLEM

Enforcement is **reactive**. Cities respond **after** the damage is done.

Illegal parking is more than a fine to be collected — it shrinks road capacity, blocks emergency routes, and snowballs into congestion. Yet most enforcement teams still operate on yesterday's complaint logs and gut feel.

01 • IMPACT



Illegal parking causes congestion

A single blocked lane on an arterial can drop effective capacity by 30–50% and amplify downstream delay across kilometres.

02 • OPERATIONS



Reactive enforcement is inefficient

Patrols dispatched after complaints arrive too late, and limited resources are spread thinly across zones of unequal severity.

03 • DATA



Raw counts hide the truth

100 violations on a wide road $\neq$ 100 on a narrow choke point. Counts ignore severity, road structure, and growth velocity.

04 • DECISION



Need prioritization & prediction

Cities need ranked hotspots, early warnings on emerging zones, and a clear next-action recommendation — not dashboards full of dots.

WHY THIS PROBLEM MATTERS

Five pressures colliding on every arterial.

These pressures don't act alone. They compound — and they need an intelligence layer that reasons across all five.



PRESSURE 01

Congestion

Illegally parked vehicles narrow lanes, slow traffic, and cascade delay into every connecting corridor.

↑ 42%

PEAK DELAY



PRESSURE 02

Blocked carriageways

Two-lane roads collapse into single-file flow. A blocked turn-lane forces unsafe merges and stalls intersections.

−50%

LANE CAPACITY



PRESSURE 03

Safety risks

Obstructed sight lines, blocked crosswalks, and emergency-vehicle delay near schools and hospitals.

CRITICAL

NEAR SENSITIVE POI



PRESSURE 04

Poor patrol allocation

Patrols rotate uniformly across beats while severity, repeat-risk and growth differ by an order of magnitude.

~70%

PATROL TIME WASTED



PRESSURE 05

No unified layer

Data, road network, traffic, vehicle history — all sit in silos. Decisions are made without joining them.

0 → 1

INTELLIGENCE LAYER

DATASET OVERVIEW

One CSV. Enough signal to power an enforcement engine.

DATASET SNAPSHOT

298,450

PARKING VIOLATION RECORDS

GEO RESOLUTION

lat / lon

point-level precision

TIME RESOLUTION

datetime

second-level

VEHICLE TRACE

vehicle #

repeat-offender linkage

EVIDENCE GATE

validated

approved-only filter

KEY COLUMNS

latitude

FLOAT • WGS84

longitude

FLOAT • WGS84

vehicle_number

STRING • plate ID

created_datetime

TIMESTAMP • IST

violation_type

CATEGORICAL

validation_status

approved / rejected

WHY THIS DATA IS ENOUGH

► Where + When + Who + What

Four pillars of an enforcement signal — all already present in one file.

► Joinable with open layers

Lat/lon enables OSMnx, Mappls, traffic — without any extra feed.

► Approved-only evidence

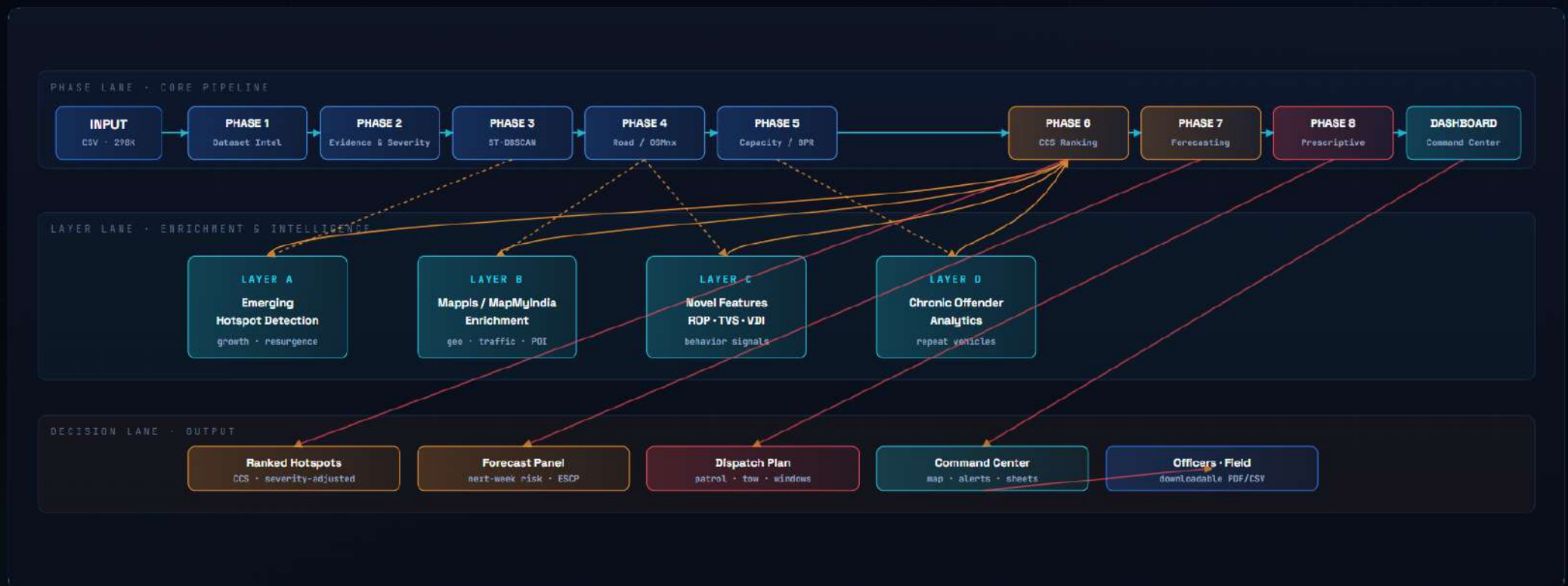
Validation flag lets us drop noise without external review pipelines.

► Volume = statistical power

~300K rows is enough to cluster, score, and forecast credibly.

END-TO-END ARCHITECTURE

From raw CSV to dispatch — one continuous intelligence pipeline.



PHASE

01

UNDERSTAND

PHASE 1 · DATASET INTELLIGENCE

Audit the data **before** you trust it.

Phase 1 is structured data understanding — validation rates, vehicle behavior, temporal rhythm and violation distribution. We surface every bias before it propagates into clustering, scoring, or forecasting.

1.1



Validation Analysis

Approved vs rejected mix, rejection reasons, reviewer drift.

APPROVED RATE

81.4%

- ▶ Reject reasons tagged
- ▶ Officer-level audit trail
- ▶ Confidence gate for downstream

1.2



Vehicle Behavior

Repeat plates, vehicle-type mix, fleet vs private signatures.

REPEAT OFFENDERS

7.2%



- ▶ Plate frequency curve
- ▶ Dominant vehicle category
- ▶ First-seen / last-seen

1.3



Temporal Analysis

Hour-of-day, weekday, peak windows, seasonal drift.



HOUR 0 → 23 (sampled) · TUE/WED/THU

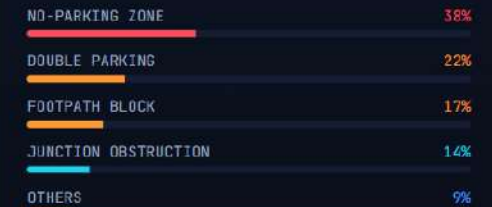
- ▶ Two-peak signature (10–13, 18–21)
- ▶ Weekend tail

1.4



Violation Analysis

Type distribution, severity mix, geographic skew.



- ▶ Type → severity mapping ready for Phase 2

PHASE

02

SCORE

PHASE 2 · EVIDENCE FILTER + SEVERITY

Not every violation hurts the road **equally**.

We keep only approved evidence, then score every record on a 1–5 severity scale tied to its real impact on lane capacity and safety. From here on, every count is weight-adjusted.

EVIDENCE GATE

RAW INPUT

All records

298,450



FILTER · validation_status = approved

Approved only

242,938



SEVERITY MAP · 1 → 5

Weighted evidence base

Each row carries a severity weight propagated into clustering and CCS.



DOWNSTREAM

Phase 3 · ST-DBSCAN

→ weighted clustering

SEVERITY MAPPING TABLE

LEVEL	VIOLATION TYPE	ROAD IMPACT	WEIGHT
5	Junction / lane-blocking obstruction	●●●●●	1.00
4	Double parking on arterial	●●●●○	0.80
3	No-parking zone	●●●○○	0.60
2	Footpath / pedestrian-edge encroachment	●●○○○	0.40
1	Minor / time-bound violations	●○○○○	0.20

Why it matters: A single Level-5 obstruction on a 2-lane arterial outweighs dozens of Level-1 violations on a wide road.

PHASE

03

CLUSTER

PHASE 3 · SPATIAL HOTSPOT DISCOVERY

From scattered dots to **named hotspots**.

ST-DBSCAN groups violations across *space and time* simultaneously — capturing real recurring hotspots, not just dense pixels on a map. Each cluster is then resolved to a named junction.

ALGORITHM · ST-DBSCAN

Spatio-Temporal DBSCAN

Density-based clustering with two distance dimensions — meters and minutes.

ESPATIAL 50 m

ETEMPORAL 60 min

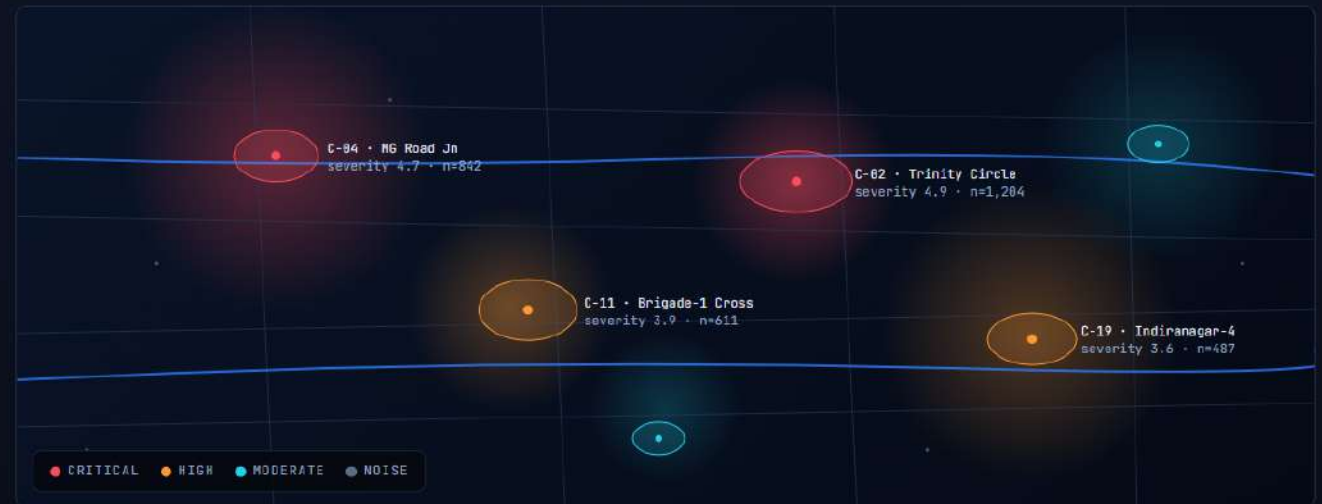
MINPTS 8

SEVERITY WEIGHT propagated

OUTPUT → cluster_id · centroid · named_junction · severity_sum

HOTSPOT CLUSTER MAP · SAMPLE ZONE

CLUSTERS = 147 · NOISE = 3.1%



LAYER

A

EARLY WARN

LAYER A · EMERGING HOTSPOT DETECTION

See the next hotspot **before** it becomes one.

Layer A monitors growth velocity, anomaly burst and resurgence across small zones — flagging neighbourhoods accelerating toward critical density, weeks before they show up in raw counts.

GROWTH TREND · 6-WEEK WINDOW

EMERGING-1 EMERGING-2 EMERGING-3 BASELINE



DETECTION SIGNALS

- Anomaly burst**
Z-score on weekly violation density per zone.
- Growth gradient**
Linear-fit slope across rolling 4-week severity sum.
- Resurgence score**
Dormant zones reigniting after enforcement gaps.

EARLY WARNING OUTPUT

23

Emerging zones flagged
CURRENT WEEK · PRE-CRITICAL

Catch them in the **amber phase** — patrol cost is lower and recurrence drops sharply when intervened early.

LAYER

B

CONTEXT

LAYER B · MAPPLS / MAPMYINDIA ENRICHMENT

Wrap every hotspot in **real-world context**.

Each cluster centroid is enriched via Mappls APIs — addresses, road class, live traffic, distance matrix, and the sensitive POIs nearby. Coordinates become decisions.

ENRICHMENT LEDGER



Reverse Geocoding

lat/lon → street, locality, junction name

[/geocode](#)

Road Class & Geometry

arterial · sub-arterial · collector + lane geometry

[/road](#)

Routing · ETA · Matrix

patrol → hotspot ETAs · nearest-unit dispatch

[/route](#)

Live Traffic Validation

does congestion match our model? confirms causality

[/traffic](#)

Sensitive POI Proximity

schools · hospitals · metro · fire stations within 200 m

[/poi](#)

ENRICHED HOTSPOT · C-62 TRINITY CIRCLE

LIVE OVERLAY



PHASE

04

NETWORK

PHASE 4 • ROAD / NETWORK INTELLIGENCE

Not all roads are **equally critical**.

We materialise the city as a road graph with OSMnx — then attach structural importance to every hotspot. A violation on a betweenness-heavy junction is a citywide problem, not a local one.

OSMnx ROAD GRAPH • BETWEENNESS HEATMAP

N = 3,418 • E = 8,286



STRUCTURAL FEATURES PER HOTSPOT

JUNCTION DEGREE

 $k = \text{deg}(v)$

how many roads meet here

BETWEENNESS

 $C_B(v)$ share of shortest paths through v

LANE COUNT

 n_{lanes}

parsed from OSM tags

CARRIAGEWAY WIDTH

 w_{CW}

used for capacity loss in Phase 5

WHY STRUCTURAL IMPORTANCE MATTERS

A hub junction with betweenness 0.94 is a **load-bearing wall** of the city's traffic graph. Blocking it propagates delay across thousands of routes. The same number of violations on a dead-end lane has near-zero downstream impact.

RULE OF THUMB → hotspot rank $\propto$ violations $\times$ severity $\times$ structural_weight

PHASE

05

MODEL

PHASE 5 · CAPACITY LOSS + CONGESTION

Translate violations into **lost seconds**.

A parked car steals lane width. Lane width drives capacity. Capacity drives delay. We close the loop with three classical models — and emit a delay-in-seconds per hotspot.

FORMULA FLOW

STEP 1 · QUEUE

$$L = \lambda / \mu$$

Average queue length from arrival rate (λ) over service rate (μ).

M/M/1 STEADY STATE



STEP 2 · CAPACITY DROP

$$C' = C \cdot f_{\text{Zhao}}(w', n)$$

Zhao reduction — effective capacity falls with shrunk lane width and obstruction count.

ZHAO ET AL · CAPACITY MODEL



STEP 3 · DELAY

$$t = t_0 [1 + \alpha(v/C')^\beta]$$

Modified BPR — travel time blows up as v/C' crosses 1.

BPR · $\alpha=0.15, \beta=4$

WHY THIS LOOP MATTERS

Parking is not the cost — **delay is**. Linking each hotspot to a measurable delay (seconds × vehicles) turns enforcement from a fines question into a **throughput** question.

AVG DELAY · TOP-20

+4m 12s

CAPACITY LOSS · PEAK

-38%

VEH-HRS LOST / WEEK

14.8k

BPR DELAY CURVE



LAYER

C

FEATURES

LAYER C • NOVEL FEATURE ENGINEERING

Five signals that capture how a hotspot behaves.

Counts say *what*. Our engineered features say *how repeat*, *how volatile*, *how diverse*, *how uncertain*, *how recurring* — the behavioural shape that drives prediction quality.

FEATURE 01

ROP

REPEAT-OFFENDER PRESSURE

Share of violations at a hotspot generated by plates seen 3+ times. High ROP → entrenched habitual misuse.

$$\text{ROP} = \sum v_{\text{repeat}} / \sum v_{\text{total}}$$

SAMPLE C-02 0.64

FEATURE 02

TVS

TEMPORAL VOLATILITY SCORE

Coefficient of variation across hourly buckets. High TVS → bursty zones that need precise time-window patrols.

$$\text{TVS} = \sigma(\lambda_h) / \mu(\lambda_h)$$

SAMPLE C-02 1.42

FEATURE 03

VDI

VIOLATION DIVERSITY INDEX

Shannon entropy across violation types at the hotspot. Diverse zones need varied enforcement playbooks.

$$\text{VDI} = -\sum p_i \cdot \log p_i$$

SAMPLE C-02 1.83

FEATURE 04

VALIDATION UNCERTAINTY

CONFIDENCE PENALTY

Hotspots with high reject ratio carry an uncertainty discount. Stops the model trusting noisy zones.

$$U = 1 - \text{approved} / \text{total}$$

SAMPLE C-02

0.11

Low uncertainty → trustworthy

FEATURE 05

RESURGENCE SCORE

COMEBACK INSTABILITY

Penalises zones that go dormant then re-ignite — the most operationally annoying pattern.

$$R = \Delta t \cdot \text{gap}_{\text{quiet}}$$

SAMPLE C-02

0.78 ↑

High recurrence

LAYER

D

VEHICLE INTEL

LAYER D • CHRONIC OFFENDER ANALYTICS

A few plates do **most of the damage**.

Vehicle-level intelligence: who is recurring, across how many hotspots, in what vehicle type, and across what time horizon. The output is a watchlist enforcement teams can act on directly.

PARETO OF OFFENCE

7.2%

OF PLATES

42%

OF SEVERITY-WEIGHTED VIOLATIONS

WATCHLIST

412 plates

flagged as chronic offenders this cycle

DOMINANT TYPE

Commercial

AVG HOTSPOTS

3.4 zones

CHRONIC OFFENDER LEDGER • TOP 8

SORT: severity-weighted count ↓

PLATE	TYPE	VIOL.	UNIQUE HOTSPOTS	FIRST SEEN	LAST SEEN	RISK
KA-01-CX-4287	Truck	82	C-02, C-04, C-11, C-19, C-23	14 Jan	22 Jun	CRITICAL
KA-05-AT-9931	LCV	74	C-02, C-07, C-19	02 Feb	21 Jun	CRITICAL
KA-03-MH-0142	Sedan	61	C-02, C-04, C-11	28 Feb	22 Jun	HIGH
KA-09-PT-6610	Mini-truck	58	C-04, C-19, C-23, C-27	11 Mar	20 Jun	HIGH
KA-02-DE-7704	SUV	47	C-11, C-23	06 Apr	19 Jun	HIGH
KA-04-SK-2218	Auto	39	C-07, C-19	19 Apr	18 Jun	MED
KA-01-RR-5380	Sedan	34	C-02	22 Apr	21 Jun	MED
KA-08-VV-9156	Truck	31	C-04, C-23	02 May	22 Jun	MED

PHASE

06

RANK

PHASE 6 · CONGESTION COST SCORE (CCS)

One number. Five forces. Honest priority.

CCS fuses delay, severity, growth, structural criticality and arrival intensity into a single rank-friendly score. It is what replaces "more dots = more dispatch."

CCS COMPOSITION

$$\text{CCS} = w_1 \cdot \text{Delay} + w_2 \cdot \text{Severity} + w_3 \cdot \text{Growth} + w_4 \cdot \text{Criticality} + w_5 \cdot \text{Arrival}$$

W₁

0.32

W₂

0.24

W₃

0.18

W₄

0.16

W₅

0.10

SCALED → each factor z-normalised, clipped, then weighted; CCS ∈ [0, 100]

RAW COUNT VS CCS · RANK FLIP

BY RAW VIOLATION COUNT

#1	C-19 Indiranagar	1,318	#1	C-02 Trinity	94.2
#2	C-02 Trinity	1,204	#2	C-04 MG Road	88.7
#3	C-04 MG Road	842	#3	C-19 Indiranagar	73.1

BY CCS (severity + impact)

TOP HOTSPOTS · CCS RANKING

CYCLE · week 25

#	HOTSPOT	DLY	SEV	GRW	CRIT	ARR	CCS
01	Trinity Circle Jn.	92	96	88	98	91	94.2
02	MG Road Jn.	85	92	76	94	88	88.7
03	Indiranagar-4	68	71	82	66	90	73.1
04	Brigade-1 Cross	63	74	62	71	66	68.9
05	Koramangala-7	58	63	79	54	61	62.4
06	Jayanagar Block-5	52	58	55	60	54	55.8
07	HSR Sector-1	48	52	60	48	49	51.3

PHASE

07

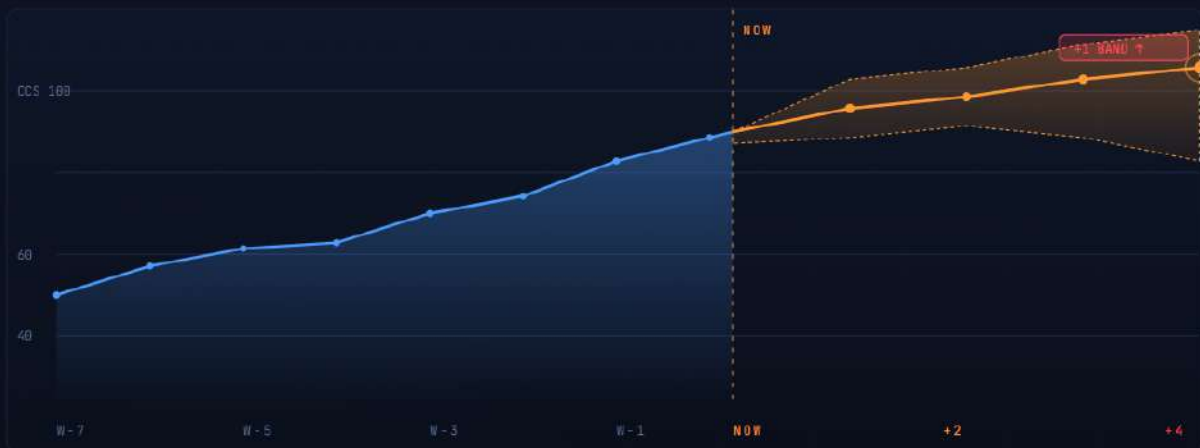
PREDICT

PHASE 7 · FORECASTING

Patrol next week, **not last week.**

An XGBoost model trained on temporal, behavioural and structural features predicts each hotspot's CCS one week ahead — plus an escalation probability that flags zones about to jump risk band.

FORECAST PANEL · C-02 TRINITY CIRCLE



MODEL · XGBoost Regressor

FEATURES

38

HORIZON

+ 1 week

MAE

4.8 pts

R²

0.87

TOP FEATURES: resurgence · CCS_{t-1} · betweenness · ROP · arrival_{peak}

ESCALATION PROBABILITY

0.84

C-02 → CRITICAL band
in next 7 days

RISK BAND PROJECTION



PHASE

08

ACT

PHASE 8 · PRESCRIPTIVE ENFORCEMENT ENGINE

From insight to **dispatch sheet**.

The engine turns ranked hotspots + forecasts into concrete operational instructions: who patrols, where, when, with what equipment, and which offenders are towed on sight.



ACTION 01

Patrol Allocation

Assign N officers to top-K CCS zones, weighted by escalation probability.

→ 28 / 32 patrols allocated



ACTION 02

Tow Readiness

Stage tow trucks pre-emptively near zones with high severity-5 historical density.

→ 6 staging points · 5 km radius



ACTION 03

Time Windows

Patrol windows derived from per-hotspot temporal signature, not uniform shift.

→ C-02: 11:00-13:30, 18:00-21:00



ACTION 04

Zone Prioritisation

Tiered into Critical / High / Watch — each tier maps to a service-level target.

→ T1 = response ≤ 8 min



ACTION 05

Offender Actions

Chronic plates trigger automatic tow / escalation flags pushed to field devices.

→ 412 plates flagged

DISPATCH PLANNER · NEXT SHIFT · 06:00 → 14:00

AUTO-GENERATED · 06:18:42

UNIT	HOTSPOT	TIER	WINDOW	ROUTE PLAN	EQUIPMENT	ETA
P-07	C-02 Trinity Circle	CRITICAL	06:30 - 09:00	Base → MG Rd → C-02 loop	+ Tow T-2 staged	6m 40s
P-12	C-04 MG Road Jn.	CRITICAL	07:00 - 10:30	Base → C-04 → C-11 sweep	+ Tow T-5 staged	8m 10s
P-19	C-19 Indiranagar-4	HIGH	08:30 - 11:00	via Brigade-1 cross	Standard kit	11m 04s
P-21	C-27 HSR Sector-1	WATCH	10:00 - 12:30	soft sweep, observation	Standard kit	14m 22s

INTERACTIVE DASHBOARD

Smart-city Command Center — every signal in one screen.



KEY RESULTS & IMPACT

What enforcement teams can now do **better**.

RANKED HOTSPOTS

147

CCS-ranked zones across the city, replacing raw-count beat lists.

→ priority is now meaningful

EMERGING HOTSPOTS

23 ↑

Pre-critical zones flagged weeks before raw counts would alert.

→ intervene in amber phase

CHRONIC OFFENDERS

412

Plates accounting for 42% of severity-weighted violations — towable on sight.

→ vehicle-level action

FORECAST ACCURACY

R² 0.87 MAE 4.8

Next-week CCS predicted within ±5 points — enables predictive patrols.

→ tomorrow's risk, today

WHAT CHANGES FOR THE ENFORCEMENT TEAM

FROM complaint-driven

→ CCS-ranked dispatch — biggest-impact zones first

FROM uniform patrols

→ Time-window-aware patrols per hotspot signature

FROM reactive

→ Predictive — next-week risk drives this-week deployment

FROM data-dashboards

→ Downloadable dispatch sheets — instructions, not graphs

EXPECTED IMPACT

2.4×

enforcement yield per patrol-hour (severity-weighted)

-38%

peak-hour capacity loss at top-20 hotspots

14.8k

vehicle-hours saved per week (modelled)

WHY THIS SOLUTION STANDS OUT

Eight properties most parking analytics tools **don't** have.

01 PIPELINE
End-to-end

Ingest → audit → cluster → enrich → score → forecast → prescribe → dispatch. One unbroken chain — no manual exports between stages.

02 EVIDENCE
Data-driven

Every output traces back to ~243K validated records. No subjective beat lists, no arbitrary thresholds, no gut-feel ranks.

03 CONTEXT
Road-aware

OSMnx betweenness, lane count and carriageway width baked into every hotspot — structural impact, not just dot density.

04 FORECAST
Predictive

Next-week CCS + escalation probability per zone (R^2 0.87). Patrol assignments lead the curve, not chase it.

05 ACTION
Prescriptive

Outputs are dispatch sheets — units, windows, routes, equipment, offender flags — not just charts to interpret.

06 DEPLOYMENT
Scalable

Same engine works for one city or fifty — only the CSV and MappIs credentials change. ST-DBSCAN and XGBoost scale linearly.

07 TRANSPARENCY
Explainable

CCS is a weighted sum — every rank can be unpacked into its five contributing factors. Auditable in court if needed.

08 PRACTICAL
Operationally useful

PDFs and CSVs that fit field workflows — not a research notebook. The officer at 06:00 reads a sheet, not a dashboard.

FUTURE SCOPE

From hackathon prototype to **live city deployment.**

Six tracks evolve the engine into a continuously-running, mobile-integrated, citywide enforcement nervous system.

 **TRACK 01**
Live deployment

Container-deploy on city ops cloud, with role-based access for control rooms, sector officers and audit teams.

 TRACK 02
Real-time feeds

Ingest CCTV / ANPR / e-challan streams. CCS refreshes every 5 minutes, not weekly.

 TRACK 03
Mobile enforcement

Officer-side app: live route, hotspot brief, offender lookup, evidence capture, e-challan integration.

 TRACK 04
Stronger forecasting

Upgrade to ST-GNN / temporal-fusion transformers; weather, events and holiday calendars as exogenous signals.

 TRACK 05
Richer validation

Vision-AI cross-check on evidence photos; severity auto-assignment with bounding-box explanations.

 TRACK 06
Dashboard automation

Auto-generated daily briefs, weekly board PDFs, and SLA-breach alerts pushed to ops over WhatsApp / Teams.

CONCLUSION

From a raw CSV to a **decision system** — parking violations become **prevented congestion**.

THE ENGINE IN ONE LINE



8 PHASES · 4 LAYERS

One engine

Ingest, model, predict, prescribe — under a single explainable score.

298,450 ROWS

Real data

Every output is traceable to validated evidence — no synthetic claims.

FIELD-READY

Dispatch, not dashboards